

ECO387C - Fall 2024

Discussion 10

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1 The McCall search model

1.1 Introduction

In all models that we have covered so far, in equilibrium, all markets clear in every period; i.e. demand and supply are always equated. However, if the labor market was truly Walrasian, then unemployment and unfilled vacancies cannot coexist, which it obviously does in reality. To account for this, economists often model labor markets with **search and matching** models instead, which does not impose market clearing.¹

Despite some obvious limitations, the McCall model's simplicity makes it extremely useful for learning dynamic programming and search and matching models. Specifically, the McCall model uses **random search**, where unemployed workers and vacancies are matched at random.

1.2 Solving the McCall model

Solving the McCall model requires us to find:

1. the value function:

$$V(w) = \max_{\{\text{Accept, Reject}\}} \left\{ \frac{w}{1 - \beta}, c + \beta \mathbb{E}(V(w')) \right\} \quad (1)$$

2. the worker's policy function:

$$\sigma(w) = \begin{cases} \text{Accept} & w \geq w^* \\ \text{Reject} & w < w^* \end{cases}$$

where w^* denotes the indifference point or **reservation wage**, the lowest wage for which the worker is indifferent between accepting or rejecting the offer.

¹Another market which often features search and matching models is the housing market.

1.3 Strategy 1: Direct attack

Note that we already know a lot about what the value function and policy function looks like. We know that the policy function is simply 0 (or reject) until the wage offer is weakly greater than w^* , at which point there is a discontinuous jump to 1 (or accept).

For the value function, we know that the value of accepting is an increasing linear function of w with slope $\frac{1}{1-\beta}$. For the value of rejecting, we know that the expectation is a constant by the iid assumption, so the entire function is a constant line that is invariant to w . The overall value function is the maximum of these two lines, and their intersection is given by w^* .

Together, this means that if we can find the reservation wage, a **scalar**, finding the value and policy function and thus solving the model becomes trivial.

Let γ denote the value associated with rejecting a wage offer:

$$\gamma = c + \beta \sum_{i=1}^N V(W_i) p_i$$

Substituting the Bellman equation (Equation 1) into the above yields:

$$\gamma = c + \beta \sum_{i=1}^N \max \left\{ \frac{w_i}{1-\beta}, \gamma \right\} p_i \quad (2)$$

Notice that the right hand side of Equation 2 is a function of γ - i.e. we simply need to find the γ such that:

$$c + \beta \sum_{i=1}^N \max \left\{ \frac{w_i}{1-\beta}, \gamma \right\} p_i - \gamma = 0$$

1.4 Strategy 2: Value function iteration

An alternative way to solve the McCall model is value function iteration. Despite being slower (we're solving for an entire vector instead of a single scalar) and less accurate, it is much more robust to various features of the model changing because it inherently relies on fewer assumptions about the environment.

See the provided code on Canvas for implementation details.

2 Leibniz's Rule

Suppose that we want to evaluate the following derivative:

$$\frac{d}{d\theta} \int_a^b f(\theta, x) dx.$$

Note that with a Taylor series approximation,

$$f(\theta + \Delta\theta, x) \approx f + \frac{df}{d\theta} \Delta\theta$$

Therefore,

$$\begin{aligned} \frac{d}{d\theta} \int_a^b f(\theta, x) dx &= \lim_{\Delta\theta \rightarrow 0} \frac{\int_a^b (f + \frac{df}{d\theta} \Delta\theta) dx - \int_a^b f dx}{\Delta\theta} \\ &= \lim_{\Delta\theta \rightarrow 0} \frac{\int_a^b f dx + \int_a^b \frac{df}{d\theta} \Delta\theta dx - \int_a^b f dx}{\Delta\theta} \\ &= \lim_{\Delta\theta \rightarrow 0} \frac{\int_a^b \frac{df}{d\theta} \Delta\theta dx}{\Delta\theta} \\ &= \int_a^b \frac{df}{d\theta} dx \end{aligned}$$

Suppose now that $f(x, \theta), a(\theta), b(\theta)$ be differentiable with respect to θ , and we want to evaluate the following derivative:

$$\frac{d}{d\theta} \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx$$

Note that using a Taylor series approximation again,

$$\frac{d}{d\theta} \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx = \lim_{\Delta\theta \rightarrow 0} \frac{\int_{a(\theta) + \frac{da}{d\theta} \Delta\theta}^{b(\theta) + \frac{db}{d\theta} \Delta\theta} (f + \frac{df}{d\theta} \Delta\theta) dx - \int_{a(\theta)}^{b(\theta)} f dx}{\Delta\theta}$$

Expanding the numerator:

$$\int_{a(\theta)}^{b(\theta)} (f + \frac{df}{d\theta} \Delta\theta) dx - \int_{a(\theta) + \frac{da}{d\theta} \Delta\theta}^{a(\theta) + \frac{da}{d\theta} \Delta\theta} (f + \frac{df}{d\theta} \Delta\theta) dx + \int_{b(\theta)}^{b(\theta) + \frac{db}{d\theta} \Delta\theta} (f + \frac{df}{d\theta} \Delta\theta) dx - \int_{a(\theta)}^{b(\theta)} f dx$$

Expanding this further, cancelling terms eventually results in:

$$\begin{aligned} \frac{d}{d\theta} \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx &= \int_{a(\theta)}^{b(\theta)} \frac{df}{d\theta} dx - \lim_{\Delta\theta \rightarrow 0} \frac{\int_{a(\theta)}^{a(\theta) + \frac{da}{d\theta} \Delta\theta} f dx}{\Delta\theta} + \lim_{\Delta\theta \rightarrow 0} \frac{\int_{b(\theta)}^{b(\theta) + \frac{db}{d\theta} \Delta\theta} f dx}{\Delta\theta} \\ &= \int_{a(\theta)}^{b(\theta)} \frac{df}{d\theta} dx - f(a(\theta), \theta) \frac{da(\theta)}{d\theta} + f(b(\theta), \theta) \frac{db(\theta)}{d\theta} \end{aligned}$$